

Table 3 Heat flux data for normalization of fin heat transfer rates

Heat flux of reference surface based on planform area, $q_b(10^{-4} \text{ W/m}^2)$	Base superheat $\Delta T_{\text{sat}} (^{\circ}\text{C})$			
	10	20	30	40
Flat reference surface (FC-72)	1.41	13.18	17.30	-
Flat reference surface (FC-87)	6.98	13.54	16.94	-
Low-profile microgroove reference surface (FC-72)	6.14	20.56	26.39	30.88
Low-profile microstud reference surface (FC-72)	9.95	24.08	29.99	36.54

ometry and may even result in effectiveness values below unity. This trend is clearly observed in Figs. 4 and 5 for large diameter fins with large aspect ratios. The high internal conduction resistances of these fins confine nucleate boiling to a very short distance from the base, leaving most of the fin surface undergoing natural convection, a condition less favorable than a bare surface undergoing boiling. Figures 4 and 5 demonstrate that, for a specific fin material at high base temperature, the effectiveness curves for various diameters initially follow a common path, while at low base superheat the curves diverge at relatively low aspect ratios.

Elemental Approach to Determining Enhanced Surface Performance. An elemental approach was employed in the calculation of effectiveness for the enhanced surfaces. Elements of surface area, A_e , were defined as identical sections of the flat surface of a chip to which a cylindrical (pin) fin is attached. Superposition of the heat transfer rate from a fin and the surrounding bare surface of an element was based on the assumption that the heat flux/wall superheat characteristics of the bare surface are insensitive to the presence of the fin. The primary advantage of this elemental approach is the ease of extension to computing the effectiveness of an entire chip covered with multiple fins. The surface was assumed to be covered with elements of identical effectiveness. One obvious limitation to this assumption is the complexity of natural convection over multi-finned surfaces and the lack of uniformity in the convection coefficient between the various elements. However, numerical results demonstrate that variations in the magnitude of the natural convection coefficient have a weak influence on the heat transfer rate from fins undergoing mostly nucleate boiling. Another limitation of the elemental analysis is the potential thickening of a vapor bubble layer on multi-finned surfaces. This problem is addressed in detail in conjunction with the experimental results of this study.

Knowing the effectiveness of the fin itself, the effectiveness and efficiency of a finned element of area A_e can be determined from the relations

$$\epsilon_e = 1 + \frac{\epsilon_f - 1}{A_e^+} \quad (8)$$

$$\eta_e = \frac{1}{A_f^+ + A_e^+ - 1} + \frac{1}{1 + \frac{A_f^+}{A_e^+ - 1}} \quad (9)$$

where $A_e^+ = A_e/A_b$.

For chips covered with m identical elements, the overall chip effectiveness is determined by the area-weighted average

$$\epsilon_{\text{chip}} = \frac{1}{A_{\text{chip}}} \sum_{i=1}^m \epsilon_{e_i} A_{e_i}, \quad i = \epsilon_e \quad (10)$$

The proper layout and sizing of elements on a chip surface

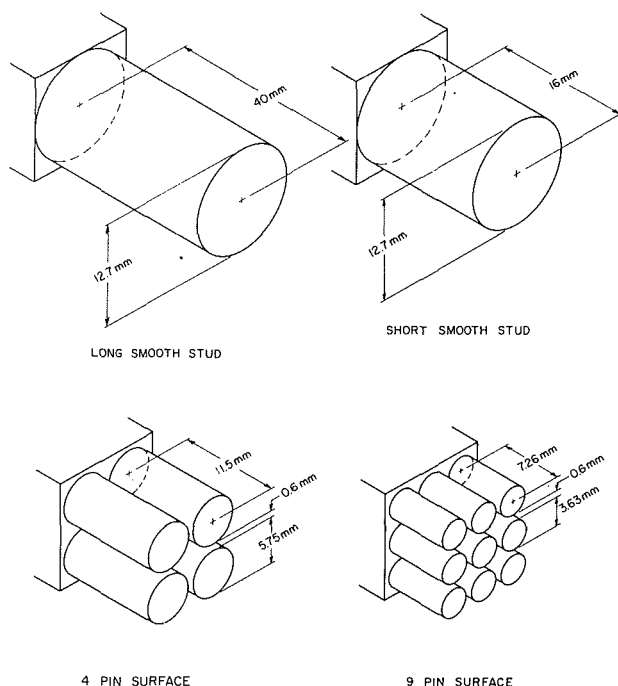


Fig. 6 Geometries of smooth studs and pin fin arrays

is limited by the possibility of severe vapor coalescence and accumulation between the fins, which may promote the formation of a vapor blanket and result in CHF. Because the critical spacing between fins is related to the generation of vapor bubbles, it is proposed that the spacing be related to the bubble departure diameter, D_b , by a scaling factor γ . Bondurant and Westwater (1971) tested radially finned horizontal tubes boiling in R-113 and found that the critical clearance between fins was in the range of the nucleate boiling departure diameter. Further experiments by Klein and Westwater (1971) using a nine pin array confirmed that a bubble departure diameter was the optimal spacing for fin arrays. Based on these findings, the scaling factor, γ , was set equal to one for the design of enhanced boiling surfaces used in the present study. Photographs obtained in the present study indicated the average D_b value for saturated boiling of FC-72 and FC-87 was approximately 0.61 mm. Accordingly, the diameter of pin fins arranged in an $m \times m$ array on a square chip of length L_{chip} was determined as

$$D = \frac{L_{\text{chip}} - mD_b}{m} \quad (11)$$

4 Experimental Results

The numerical codes used to generate the effectiveness plots were modified to output boiling curves based on the geometries of several enhanced surface structures used in the experimental study. The numerical predictions estimate boiling performance up to the CHF point corresponding to the flat, microgroove or microstud reference surfaces. The generalized nature of the numerical scheme allows much flexibility in the specification of design envelope. For the present study, various metallic attachments were designed to enhance heat transfer from a $12.7 \times 12.7 \text{ mm}^2$ simulated microelectronic heat source.

4.1 Pin-Fin Surface Geometries. The experimental enhancement surfaces were based on cylindrical shapes fabricated from high purity copper and arranged either as single studs or in pin arrays. Each of the enhanced surfaces was vapor blasted with an air/water/silica slurry to ensure uniform surface characteristics (Anderson and Mudawar, 1989). The two smooth stud enhanced surfaces, Fig. 6, differed only in length. The